Advanced SQL Server Performance Tuning

A two-day course in real-world performance troubleshooting on SQL Server 2025. It builds on course 65317A and teaches teams to tune and optimize queries using native SQL Server features and SSMS 22.

DURATION	CONTENT	LEVEL	DELIVERY
2 Days	6 Modules	Advanced	Instructor-Led · Bootcamp · Self-Paced

WHAT SETS THIS COURSE APART

01

Updated for SQL Server 2025

Troubleshooting is taught against SQL Server 2025 using SSMS 22 and native SQL Server features.

- Query Store diagnostics in SQL Server 2025
- Adaptive query processing in SQL Server 2025
- SSMS 22 diagnostic features

02

AI Tutor, available 24/7

An interactive AI assistant is built into the courseware platform, at no additional cost.

- Answers questions in real time
- Explains execution plans, indexing and Query Store telemetry
- Covers all other course topics

03

Lab simulations, at the student's pace

Students step back and forth through each lab exercise, revisiting any step as needed.

- Runs in a browser, with no setup required
- Replay any time, including after class
- Covers the lab in every module

ALSO INCLUDED

Bootcamp mode	One capstone lab per day. Students apply that day's skills to a new scenario with no procedure and no hints.
Methodology	A systematic troubleshooting workflow built on performance baselines.
Bottlenecks	Wait statistics used to diagnose CPU, I/O and memory bottlenecks.
Concurrency	Blocking chains, deadlocks and the impact of isolation levels.

CURRICULUM AT A GLANCE

DAY 1	**Find the problem.** Performance troubleshooting methodology, advanced index tuning, and Query Store performance diagnostics.
DAY 2	**Fix the problem.** Wait statistics and resource bottlenecks, concurrency and blocking, and advanced query optimization, including parameter sniffing, plan stability and memory grants.

What your team leaves with

The ability to analyze execution plans to identify bottlenecks, apply real-world tuning techniques with SSMS 22 and advanced diagnostic tools, optimize queries through index strategy, resolve performance issues in production environments, and use SQL Server 2025 features and Microsoft tools for continuous performance monitoring.

Prerequisites: Course 65317A, Querying and Analyzing SQL Server 2025, or equivalent knowledge; a solid understanding of T-SQL and query fundamentals; and experience with SQL Server in production environments.

COURSEWARE ACCESS

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